PY-EXPO 2026

Team Name - SIGNFLOW

Problem Statement ID - PY072

Problem Statement Title - ASL TO ENGLISH

Domain- AIML

PS Category- Software

Industry Mentor - Mathi Yuvarajan

Student Mentor - Arunaw Rishe

TEAM DETAILS

S.No.	DEPARTMENT	ROLL NO	Role	NAME
1	CYS	25UCY127	TL,Media PipeLine	Mano R P
2	CYS	25UCY115	UI/UX	Hari Varna V S
3	CYS	25UCY131	Model Training	Nithin G
4	AIDS	25UAD121	Architecture and System Designing	Paul Raja I
5	CSE	25UCS107	R&D	Amritha N
6	AIML	25UAM212	DataSet	Nithish A
7	Arunaw Rishe			

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Proposed Solution:

AI-based system for real-time translation between Sign ↔ Text ↔ Speech using computer vision and deep learning.

Works via camera AND SCREEN CAPTURING .

Innovation:

- Bi-directional translation
- Web-based & scalable
- Supports regional sign languages

How the Issue Is Solved:

- Detects hand gestures using AI (OpenCV + TensorFlow)
- Converts gestures → text/speech
- Converts speech → sign animations
- Enables instant communication without interpreters

Creativity & Feasibility:

MVP:

- Web app + live camera
- Pre-trained gesture model
- Speech/Text conversion

Benefits:

- Improves accessibility
- Fast & cost-effective
- Easy to deploy

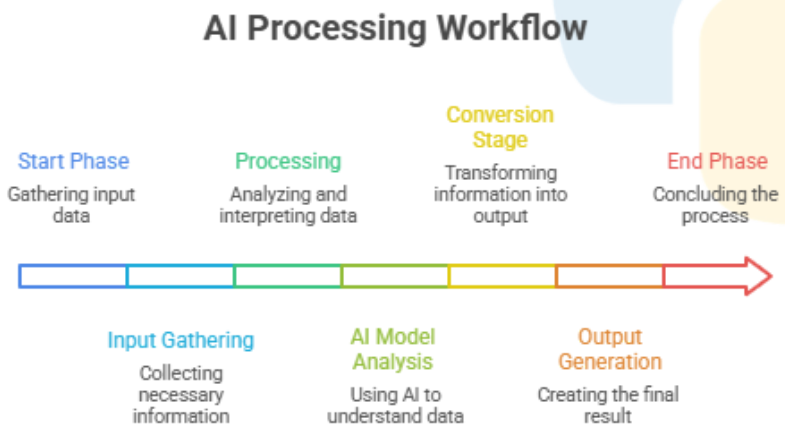
Applications:

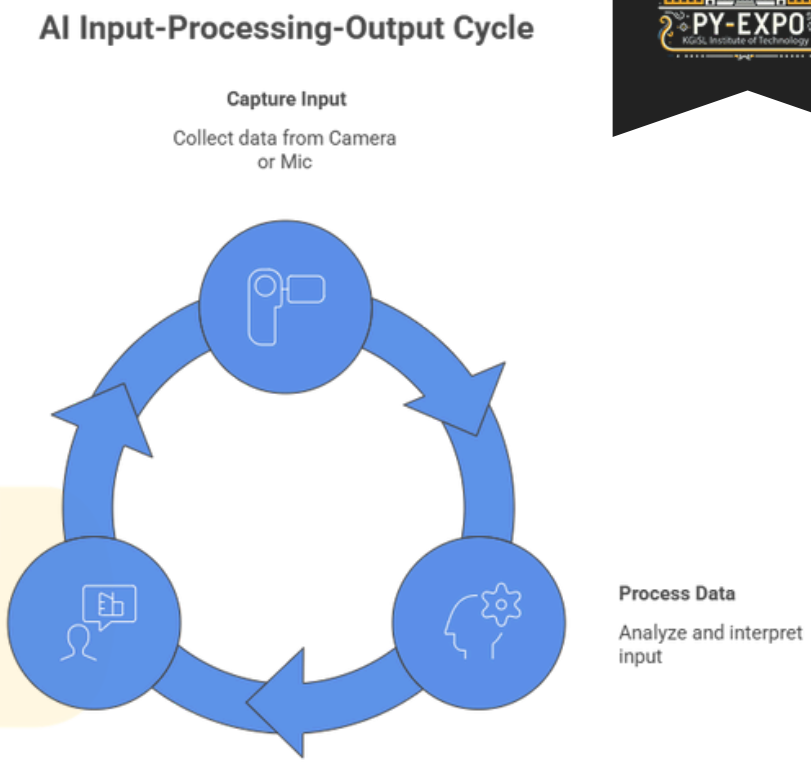
- Healthcare
- Education
- Public services

TECHNICAL APPROACH

Technical Stack:

- Frontend:Pyqt5
- Backend: Python, ollama(grammar correction)
- AI/ML: Pytorch - 3d landmarks transformer
- Computer Vision: OpenCV, mediapipe





Flow / Methodology:

- Capture input (gesture/speech)
- Preprocess data
- Apply trained AI model
- Convert to required format
- Display output in real-time

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- **Git hub -**
- **Application - Not Deployed yet**